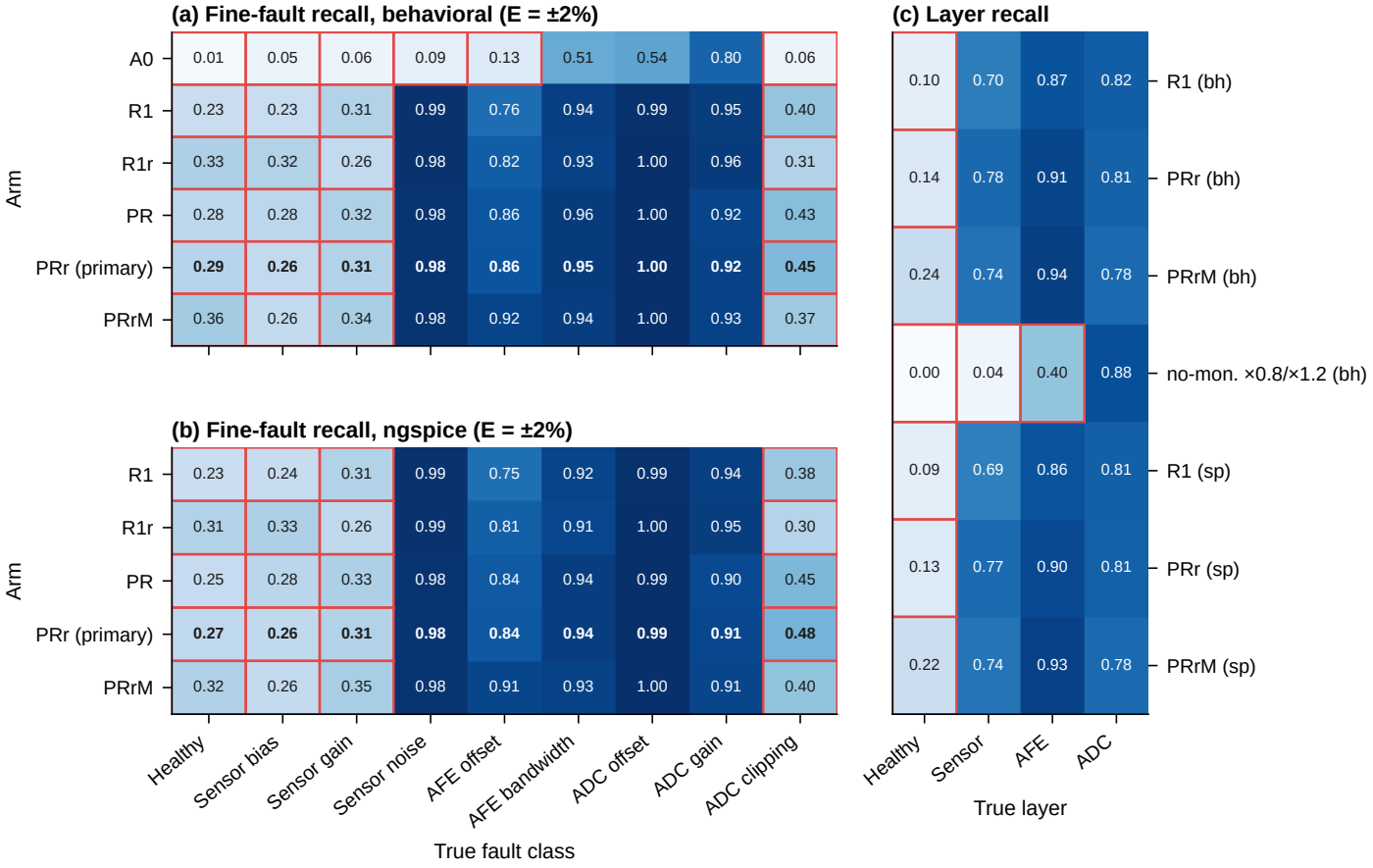

(a) Fine-fault recall, behavioral (E = ±2%)
A0 0.01 0.05 0.06 0.09 0.13 0.51 0.54 0.80 0.06
R1 0.23 0.23 0.31 0.99 0.76 0.94 0.99 0.95 0.40
R1r 0.33 0.32 0.26 0.98 0.82 0.93 1.00 0.96 0.31
PR 0.28 0.28 0.32 0.98 0.86 0.96 1.00 0.92 0.43
PRr (primary) 0.29 0.26 0.31 0.98 0.86 0.95 1.00 0.92 0.45
PRrM 0.36 0.26 0.34 0.98 0.92 0.94 1.00 0.93 0.37
Arm
(b) Fine-fault recall, ngspice (E = ±2%)
R1 0.23 0.24 0.31 0.99 0.75 0.92 0.99 0.94 0.38
R1r 0.31 0.33 0.26 0.99 0.81 0.91 1.00 0.95 0.30
PR 0.25 0.28 0.33 0.98 0.84 0.94 0.99 0.90 0.45
PRr (primary) 0.27 0.26 0.31 0.98 0.84 0.94 0.99 0.91 0.48
PRrM 0.32 0.26 0.35 0.98 0.91 0.93 1.00 0.91 0.40
Arm
Healthy
Sensor bias
Sensor gain
Sensor noise
AFE offset
AFE bandwidth
ADC offset
ADC gain
ADC clipping
True fault class
(c) Layer recall
0.10 0.70 0.87 0.82 R1 (bh)
0.14 0.78 0.91 0.81 PRr (bh)
0.24 0.74 0.94 0.78 PRrM (bh)
0.00 0.04 0.40 0.88 no-mon. ×0.8/×1.2 (bh)
0.09 0.69 0.86 0.81 R1 (sp)
0.13 0.77 0.90 0.81 PRr (sp)
0.22 0.74 0.93 0.78 PRrM (sp)
Healthy
Sensor
AFE
ADC
True layer